

# Kanghoon Yoon

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## Education

### Korea Advanced Institute of Science & Technology (KAIST)

*Daejeon, South Korea*

Ph.D. in Industrial & Systems Engineering

*Sep. 2021 – Aug. 2025*

- Advisor: Prof. Chanyoung Park
- Dissertation: Advancing Deep Neural Networks for Graph-Structured Data: Enhancing Representation Learning and Robustness to Data Bias

### Korea Advanced Institute of Science & Technology (KAIST)

*Daejeon, South Korea*

M.S. in Industrial & Systems Engineering

*Sep. 2019 – Aug. 2021*

- Advisor: Prof. Jinkyoo Park
- Dissertation: Learning Multivariate Hawkes Process using Graph Recurrent Neural Network

### Hanyang University

*Seoul, South Korea*

B.S. in Mathematics

*Mar. 2013 – Feb. 2019*

- Scholarship: Hanyang Brain Award (2017 Fall, 2018 Spring)
- Completed military service (2015 – 2017)

## Work Experience

### HUAWEI Norbert Wiener Research Center (Visa sponsored via Persolkelly)

*Singapore*

Senior AI Researcher @ Processor Algorithm Lab

*Nov. 2025 – Present*

- Developing latent-action-based VLA models to achieve precise single-/multi-arm robotic control.
- Designing a quantized post-training method to reduce the memory usage (FP8 model params, and FP8 rollout).
- Built inference engine of one trillion MoE models (Ling2.5-1T) using CANN on Ascend NPUs.

### NAVER Cloud

*Gyeonggi-do, South Korea*

Research intern @ Efficient AI Team

*Feb. 2025 – Aug. 2025*

- Accelerated inference of LLMs by 1.3x through a relaxed token verification paradigm, ensuring semantic coherence [C17].

### Qualcomm AI Research

*San Diego, CA*

Research intern @ AI System Team

*Jun. 2024 – Sep. 2024*

- Built a retrieval-based speculative decoding method using suffix-tree-based prefix matching, achieving 4.5x speedup for domain-specific LLM agents such as travel planning and tool calling.

## Research Interest

### World Models & Physical Scene Understanding

*Aug. 2021 – Present*

- Learning structured, relational representations of the physical world from vision — scene graphs as compositional world representations — and extending them to dynamic, temporal understanding in video toward models that capture how the world evolves [C2, C6, C8, C10, C12, C13].
- Action prediction and vision-language-action models for embodied agents across heterogeneous hardware and unseen environments using world knowledge.

### Efficient & On-Device Foundation Models

*Jun. 2024 – Present*

- Bringing large foundation models to the physical world: real-time, low-latency inference of Vision-Language-Action, Omni, and MoE models on edge accelerators (GPU/NPU), enabling world models to run on-device for robotic control.
- Model compression via quantization for generative foundation and VLA models, and inference acceleration through speculative and relaxed-token-verification decoding [C17].

## Research Project

### (P3) ETRI - Developing Visual Intelligence Memory via Scene Graph Generation

*Daejeon, South Korea*

Student Researcher (Lead)

*09. 2021 – 12. 2024*

- Spearheaded the development of a scene understanding algorithm to mitigate prediction bias, publishing three papers (C2, C6, C8) in top-tier conferences.

## **(P2) Shinhan Card - Personalized Store Coupon Issuance & Recommendation System**

Seoul, South Korea

Student Researcher (Lead)

09. 2020 — 03. 2021

- Architected and implemented a scalable, deep-learning-based recommendation engine for personalized coupon issuance, enhancing user engagement.

## **(P1) Shinhan Card - Personalized User Analysis via Latent Representation Clustering**

Seoul, South Korea

Student Researcher

12. 2019 — 02. 2020

- Engineered a user analysis framework utilizing latent representation clustering to identify and segment distinct user behaviors.

## **Publications**

### **CONFERENCES**

\* represents the equally-contributed authors

#### **(C17) SelfJudge: Faster Speculative Decoding via Self-Supervised Judge Verification.**

ICML 2026

Kanghoon Yoon, Minsub Kim, Sungjae Lee, Joonhyung Lee, S.H Woo, Yeonjun In, S.J Kwon, Chanyoung Park, Dongsoo Lee

#### **(C16) Image is All You Need: Towards Efficient and Effective Large Language Model-Based Recommender Systems.**

ICLR 2026

Kibum Kim, Sein Kim, Hongseok Kang, Jiwan Kim, Heewong Noh, Yeonjun In, Kanghoon Yoon, Jinoh Oh, Chanyoung Park.

#### **(C15) Training Robust Graph Neural Networks by Modeling Noise Dependencies.**

NeurIPS 2025

Yeonjun In, Kanghoon Yoon, Sukwon Yun, Kibum Kim, Sungchul Kim, Chanyoung Park.

#### **(C14) Is Safety Standard Same for Everyone? User-Specific Safety Evaluation of Large Language Models.**

EMNLP 2025  
(Findings)

Yeonjun In, Wonjoong Kim, Kanghoon Yoon, Sein Kim, M Tanjim, Kibum Kim, Jinoh Oh, Chanyoung Park.

#### **(C13) Weakly Supervised Video Scene Graph Generation via Natural Language Supervision .**

ICLR 2025

Kibum Kim, Kanghoon Yoon, Jaehyeong Jeon, Yeonjun In, Jinyoung Moon, Donghyun Kim, Chanyoung Park.

#### **(C12) Retrieval-Augmented Scene Graph Generation via Multi-Prototype Learning.**

AAAI 2025

Kanghoon Yoon, Kibum Kim, Jaehyeong Jeon, Yeonjun In, Donghyun Kim, Chanyoung Park.

#### **(C11) Revisiting Fake News Detection: Towards Temporality-aware Evaluation by Leveraging Engagement Earliness.**

WSDM 2025  
(Oral)

Junghoon Kim, Junmo Lee, Yeonjun In, Kanghoon Yoon, Chanyoung Park.

#### **(C10) Semantic Diversity-aware Prototype-based Learning for Unbiased Scene Graph Generation.**

ECCV 2024

Jaehyeong Jeon, Kibum Kim, Kanghoon Yoon, Chanyoung Park.

#### **(C9) Debiased Graph Poisoning Attack via Contrastive Surrogate Objective**

CIKM 2024

Kanghoon Yoon, Yeonjun In, Namkyeong Lee, Kibum Kim, Chanyoung Park.

#### **(C8) LLM4SGG: Large Language Model for Weakly Supervised Scene Graph Generation.**

CVPR 2024

Kibum Kim, Kanghoon Yoon, Jaehyeong Jeon, Yeonjun In, Jinyoung Moon, Donghyun Kim, Chanyoung Park.

#### **(C7) Self-guided Robust Graph Structure Refinement.**

WWW'24 (Oral)

Yeonjun In, Kanghoon Yoon, Kibum Kim, Kijung Shin, Chanyoung Park.

#### **(C6) Adaptive Self-training Framework for Fine-grained Scene Graph Generation.**

ICLR 2024

Kibum Kim\*, Kanghoon Yoon\*, Yeonjun In, Jinyoung Moon, Donghyun Kim, Chanyoung Park.

#### **(S1) Class Label-aware Graph Anomaly Detection.**

CIKM'23 (Short)

Junghoon Kim, Yeonjun In, Kanghoon Yoon, Junmo Lee, Chanyoung Park.

#### **(C5) Similarity Preserving Adversarial Graph Contrastive Learning.**

KDD'23

Yeonjun In\*, Kanghoon Yoon\*. Chanyoung Park

#### **(C4) Shift-Robust Molecular Relational Learning with Causal Substructure.**

KDD'23

Namkyeong Lee, Kanghoon Yoon. Gyoung S. Na, Sein Kim, Chanyoung Park

#### **(C3) Learning Multivariate Hawkes Process via Graph Recurrent Neural Network.**

KDD'23

Kanghoon Yoon\*. Youngjun Im\*. Jingyu Choi, Taehwan Jeong, Jinkyoo Park.

#### **(C2) Unbiased Heterogeneous Scene Graph Generation with Relation-aware Message Passing Neural Network.**

AAAI 2023

Kanghoon Yoon\*. Kibum Kim\*. Jinyoung Moon. Chanyoung Park.

## **(C1) LTE4G: Long-Tail Experts for Graph Neural Networks.**

CIKM 2022

Sukwon Yun, Kibum Ki, Kanghoon Yoon, Chanyoung Park

## **JOURNALS**

\* represents the equally-contributed authors

## **(J1) A New DAWN for Fake News Detection: Leveraging Engagement Earliness for Temporality-aware Evaluation.**

ACM TIST 2026

Junghoon Kim\*, Junmo Lee\*, Yeonjun In, Kanghoon Yoon, Chanyoung Park.

## **Invited Talks**

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### **Robust Graph Contrastive Learning**

Busan, South Korea

Korea Software Congress

12. 2023

### **Heterogeneous Scene Graph Generation**

Jeju, South Korea

Korea Computer Congress

06. 2023

## **Awards**

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### **Excellence Award in Poster Competition (2022, 2023)**

KAIST ISysE, 2022-2023

### **Hanyang Brain Scholarship**

Hanyang University, 2017-2018

## **Services**

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### **Area Chair of International Conferences**

KDD (2025-2)

### **Reviewer of International Conferences**

NeurIPS (2025), ICML (2026), ICLR (2026), CVPR (2026), AAAI (2024, 2025, 2026), KDD (2024-1, 2024-2, 2025-1)

### **Reviewer of International Journals**

TKDD (2024), TPAMI (2024, 2025)

### **Teaching Assistant**

KAIST Micro Degree (2024), ML Basics @ Hankook Tire (2024), Data Science Competition @ KAIST (2023), IE241-Engineering Statistics (2020), LG Academy Teaching (2019-2021)